

3.3.13 Threats, Stalking, Harassment, and Terrorism

Lower Bound Computation

$$\begin{bmatrix} \mathbf{x}_{\text{American Samoa}}^{(\text{count})} \\ \mathbf{x}_{\text{American Samoa}}^{(\text{dollars})} \end{bmatrix} = \begin{bmatrix} 3 & 3 & \cdots & 1 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\begin{bmatrix} \mathbf{x}_{\text{American Samoa}}^{(\text{count scaled to sample population})} \\ \mathbf{x}_{\text{American Samoa}}^{(\text{dollars scaled to sample population})} \end{bmatrix} = \begin{bmatrix} 10 & 10 & \cdots & 3 \\ \$0 & \$0 & \cdots & \$0 \end{bmatrix}.$$

$$\bar{X}_{\text{American Samoa}}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{\text{American Samoa}, i}^{(m)},$$

$$m \in \{\text{scaled count, scaled dollars}\}$$

$$\begin{bmatrix} \bar{X}_{\text{American Samoa}}^{(\text{scaled count})} \\ \bar{X}_{\text{American Samoa}}^{(\text{scaled dollars})} \end{bmatrix} = \begin{bmatrix} 7 \\ \$0 \end{bmatrix}.$$

Upper Bound Computation

$$\begin{bmatrix} \mathbf{x}_{\text{California}}^{(\text{count})} \\ \mathbf{x}_{\text{California}}^{(\text{dollars})} \end{bmatrix} = \begin{bmatrix} 2,229 & 2,441 & \cdots & 295 \\ \$2,139,776 & \$1,578,405 & \cdots & \$421,261 \end{bmatrix}.$$

$$\begin{bmatrix} \mathbf{x}_{\text{California}}^{(\text{count scaled to sample population})} \\ \mathbf{x}_{\text{California}}^{(\text{dollars scaled to sample population})} \end{bmatrix} = \begin{bmatrix} 10 & 11 & \cdots & 1 \\ \$9,599 & \$7,112.85 & \cdots & \$1,428 \end{bmatrix}.$$

$$\bar{X}_{\text{California}}^{(m)} = \frac{1}{N} \sum_{i=1}^N X_{\text{California}, i}^{(m)},$$

$$m \in \{\text{scaled count, scaled dollars}\}$$

$$\begin{bmatrix} \bar{X}_{\text{California}}^{(\text{scaled count})} \\ \bar{X}_{\text{California}}^{(\text{scaled dollars})} \end{bmatrix} = \begin{bmatrix} 11 \\ \$7,954 \end{bmatrix}.$$

Monte Carlo Simulations for Likelihood and Impact

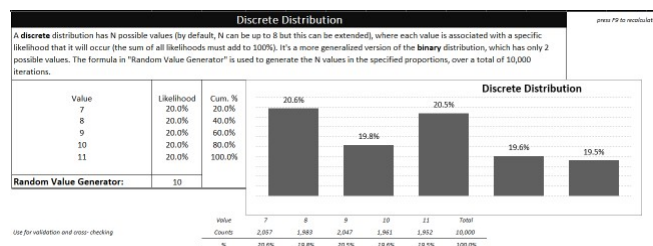


Fig. 3.35: We ran a Monte Carlo simulation of 10,000 iterations with a discrete distribution to spread random instances between confidence intervals taken from the means of seven year samples. We bounded between 7 and 11 possible threat (threats, stalking, harrassment, and terrorism) events over a year.[107] [108]

Table 3.6: Monte Carlo Discrete Simulation Values for Threats, Stalking, Harrassment, and Terrorism

Events	7	8	9	10	11
Likelihood	20.6%	19.8%	20.5%	19.6%	19.5%

A Monte Carlo simulation was run using a discrete distribution. We bounded the simulation between 7 and 11 events. We assigned a 20% likelihood to each bin. The resulting distribution showed the effect of randomness on each possible outcome.

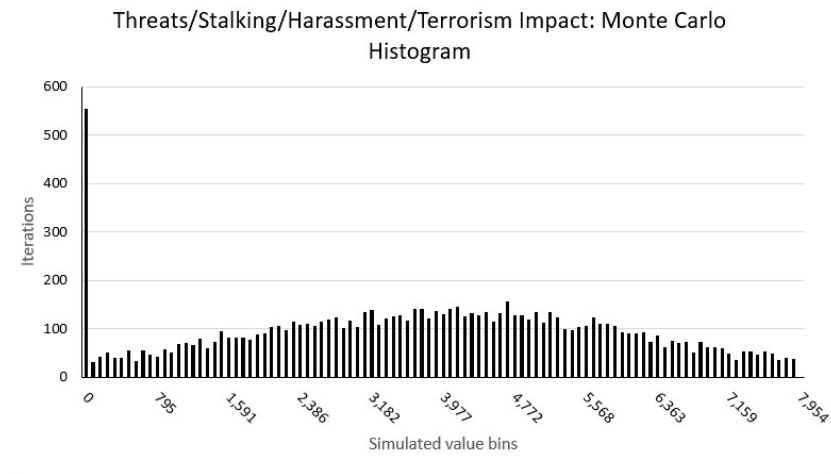


Fig. 3.36: We ran a Monte Carlo simulation of 10,000 iterations to spread random instances between confidence intervals taken from means of a seven year sample. We set a lower bound that the event would cost at \$0. We set an upper bound that the event would cost \$7,954.[109] [110]

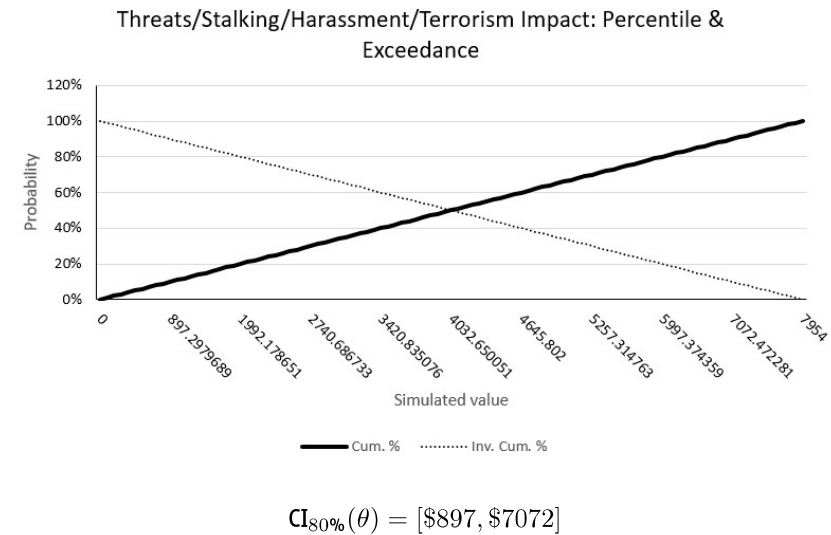


Fig. 3.37: A Monte Carlo simulation suggests that there is a 90% probability that threat events will cost \$897.29 or more in a year. There is a 50% likelihood that they will cost \$4,032.65 or more in a year. There is a 10% chance that they will cost \$7,072.47 or more in a year. [111] [112]